The Missing Component of Permanent Risk

Job Market Paper

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Goal of the paper

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2. **Challenge**: in leading AP models, permanent component of the SDF comes almost entirely from shocks related to the permanent component of growth
3. **Theory**: OLG with organization capital that generates the missing component
4. **Quantitative and empirical**
   - Two factors: growth and organization capital span the permanent risk
   - Test unique implications of the model

Permanent component of the SDF (Permanent Risk)

- Permanent component is the most important component of the SDF (Alvarez and Jermann, 2005, ECMA)

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An upper bound on the explained variation

- PC of growth accounts for at most 35% of PC of SDF

Solution

A GE model with incomplete markets and organization capital

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    - Existing investors are the ones who price the assets

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1. OLG with limited inter-cohort risk sharing
   - Existing investors are the ones who price the assets
2. Existing cohorts of firms/investors face permanent shocks to organization capital $\Rightarrow$ PC of the SDF
3. Organization capital specific shocks do not affect future organization capital $\Rightarrow$ separation of PC of SDF and PC of growth

Related literature

- **Long-term pricing**: Alvarez and Jermann (2005), Hansen and Scheinkman (2009), Bakshi and Chabi-Yo (2012), Borovička, Hansen and Scheinkman (2016), Ai and Fu (2026a, b)
- **Equity term structure**: Binsbergen and Koijen (2017), Gonçalves (2021), Bansal, Miller, Song and Yaron (2021), Belo, Collin-Dufresne and Goldstein (2015), Binsbergen (2025)
- **Organization capital**: Prescott and Visscher (1980), Atkeson and Kehoe (2005), Eisfeldt and Papanikolaou (2013)
- **OLG and limited intergenerational risk sharing**: Gârleanu, Kogan and Panageas (2012), Kogan, Papanikolaou and Stoffman (2020), Gârleanu and Panageas (2023)

Roadmap

Permanent Risk Puzzle

OLG with Organization Capital Risk: Endowment-based

OLG with Organization Capital Risk: GE with Production

Quantitative Implication and Empirical Test

Conclusion

Permanent Risk Puzzle

Long-term factorization (Hansen and Scheinkman, 2009)

- Pricing kernel S_t and consumption Y_t:

$$S_t = \underbrace{e^{-r_S t}}_{\text{trend}} \underbrace{f_S(x_t)}_{\text{transitory}} \underbrace{\hat{S}_t}_{\text{permanent}} \qquad Y_t = \underbrace{e^{\rho_Y t}}_{\text{trend}} \underbrace{f_Y(x_t)}_{\text{transitory}} \underbrace{\hat{Y}_t}_{\text{permanent}}$$

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- Alvarez and Jermann (2005) find sizable variation in $\hat{S}_t$

$$\log \frac{\hat{S}_{t+1}}{\hat{S}_t} = \log \frac{S_{t+1}}{S_t} + r_B^\infty$$

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- In AP models, they are almost perfectly correlated: $\rho\left(\log \frac{\hat{S}_{t+1}}{\hat{S}_t}, r_Y^\infty - r_B^\infty\right) \approx 1$

Example: CRRA utility

- Consumption with permanent and transitory components

$$\ln Y_{t+1} - \ln Y_t = \underbrace{\mu_C + \sigma_\epsilon \epsilon_{t+1}}_{\text{random walk}} + \underbrace{f_Y(x_{t+1}) - f_Y(x_t)}_{\text{transitory}}$$

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$$S_t = \beta^t \, Y_t^{-\gamma}$$

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$$\log \frac{\hat{S}_{t+1}}{\hat{S}_t} = const - \gamma \left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1} \right)$$

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- How about leading AP models?

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- What is the R^2 of this regression? $\Rightarrow$ $\mathbf{R^2 = 1}$
- How about leading AP models? $\Rightarrow$ $\mathbf{R^2 \approx 1}$ (a few slides later)

Permanent Risk Puzzle

Linear projection of the log permanent component of the SDF:

$$\log \frac{\hat{S}_{t+1}}{\hat{S}_t} = \alpha + \beta \left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1} \right) + u_{t+1}$$

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Proposition: An upper bound on R^2

Under log normality, for any asset i

$$R^2 \leq \frac{\left[E\left(r^{\infty}_{Y} - r^{\infty}_{B} \right) + \frac{1}{2} \mathrm{Var}\left(r^{\infty}_{Y} - r^{\infty}_{B} \right) \right]^2}{\mathrm{Var}\left(r^{\infty}_{Y} - r^{\infty}_{B} \right) \times 2E\left(r_i - r^{\infty}_{B} \right)}$$

Log Sharpe ratio over long bond

Definition

1) Log Sharpe ratio over the long-term bond of asset i

$$\Lambda_i = \frac{E\left(r_i - r_B^\infty\right) + \frac{1}{2}\sigma^2\left(r_i - r_B^\infty\right)}{\sigma\left(r_i - r_B^\infty\right)}$$

2) Optimal growth (OG) portfolio: the one with the largest expected log return

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Proposition

Under log normality, for any asset i

$$\Lambda_i \leq \sigma(\hat{s}) = \Lambda_{max}$$

Denote Λ_∞ as the ratio for the long-term equity

$$R^2 = \left(\frac{\Lambda_\infty}{\Lambda_{max}}\right)^2 = \left(\frac{\Lambda_\infty}{\Lambda_{OG}}\right)^2$$

Permanent Risk Puzzle: data

Proxy for long-term equity:

- A portfolio of firms that never paid a dividend before time t
  - 82% of them never pay a dividend in the entire sample
  - Similar to a portfolio of firms that state in their 10-K that they will not pay dividends

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Proxy for the optimal growth portfolio:

- Levered market: market and T-bill weighted to maximize the log portfolio return

Permanent Risk Puzzle: data

Upper bound on R^2	Long-term equity ($\Lambda_\infty = 0.224$)	Top-decile growth ($\Lambda_\infty = 0.263$)
Levered market ($\Lambda_i = 0.382$)	34.4	47.5

Leading asset pricing models

	Data	Habit CC (1999)	Long-run risk BY (2004)	Disaster Barro (2006)	Intermediary HK (2013)
Panel A					
$L\left(\frac{\hat{S}_{t+1}}{\hat{S}_t}\right)$	≥ 7.1	13.8	18.1	8.6	1.6

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Panel B					
$E\left[r_Y^\infty - r_B^\infty\right]$	2.5	13.8	3.8	5.2	1.2
$\sigma\left(r_Y^\infty - r_B^\infty\right)$	25.0	51.3	6.9	5.9	9.0
Λ_∞	0.224	0.526	0.586	0.922	0.180

OLG with Organization Capital Risk: Endowment-based

Technologies

Frontier Technology

- Blueprints and methods that any firm can adopt

$$\Delta \ln Z_{t+1} = \mu_z + x_{z,t}, \qquad x_{z,t+1} = \rho_z x_{z,t} + \sigma_z \epsilon_{z,t+1}$$

Organization Capital

- Firm-specific, lost when the firm exits
- If firm i has $o_{i,t}$, then it grows to $o_{i,t+1} = \frac{A_{t+1}}{A_t} o_{i,t}$
- Organization Capital Growth Index A_t:

$$\Delta \ln A_{t+1} = x_{o,t} + \sigma_\eta \eta_{o,t+1}, \qquad x_{o,t+1} = \rho_o x_{o,t} + \sigma_o \epsilon_{o,t+1}$$

ϵ_z, ϵ_o, η_o: i.i.d. shocks

Endowment

Demographics

- A unit mass of agents
- Every period, a mass κ enters and a mass κ dies

Endowment

- Firms of cohort s are endowed with organization capital $o_{s,s} = x_0$
- Organization capital of each firm grows with the index A_t: $o_{s,t+1} = \frac{A_{t+1}}{A_t} o_{s,t}$
- A firm of cohort s produces output $y_{s,t} = Z_t o_{s,t}$

Aggregation

- Aggregate output: summing over the output of all cohorts

$$Y_t = Z_t \sum_{s \leq t} \kappa (1-\kappa)^{t-s} o_{s,t} = Z_t \, O_t$$

- Denote O_t as aggregate organization capital. Its law of motion:

$$O_t = \underbrace{\kappa x_0}_{\text{entrants}} + \underbrace{(1-\kappa) \frac{A_t}{A_{t-1}} O_{t-1}}_{\text{surviving incumbents}}$$

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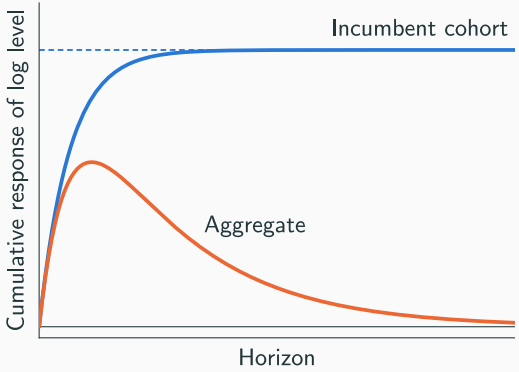
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- **Proposition (cointegration)**: if κ is large enough, then O_t is stationary and $\ln Y_t$, $\ln Z_t$ are cointegrated

Permanent for cohorts, transitory in aggregate

Response to a unit shock to x_o



- Recursive utility over own consumption C_t^s

$$U_t^s = \left[(1-\beta) \left(C_t^s \right)^{1-\frac{1}{\psi}} + \beta \left(1-\kappa \right) \left(E_t \left[\left(U_{t+1}^s \right)^{1-\gamma} \right] \right)^{\frac{1-1/\psi}{1-\gamma}} \right]^{\frac{1}{1-1/\psi}}$$

- Existing cohorts trade a complete set of Arrow securities, but cannot trade with unborn cohorts. At every time t, the newly born cohort s joins the market
- Lifetime budget constraint of cohort s

$$E_s \sum_{a \geq 0} (1-\kappa)^a \frac{S_{s+a}}{S_s} C_{s+a}^s \;\leq\; E_s \sum_{a \geq 0} (1-\kappa)^a \frac{S_{s+a}}{S_s} o_{s,s+a} Z_{s+a}$$

Equilibrium

Competitive Equilibrium

An SDF $\{S_t\}$ and consumption plans $\{C_t^s\}$ such that

1. Every cohort maximizes U_s^s subject to its lifetime budget constraint;
2. Goods market clears: $\sum_{s \leq t} \kappa (1-\kappa)^{t-s} C_t^s = Y_t$

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Equilibrium Allocation is Autarkic in this Baseline Economy

- Everyone consumes their own endowment: $C_t^s = o_{s,t} Z_t$ with growth rate $g_{t+1} = \Delta \ln Z_{t+1} + \Delta \ln A_{t+1}$
- Equilibrium SDF follows

$$\frac{S_{t+1}}{S_t} = \beta \, e^{-\frac{1}{\psi} g_{t+1}} \left(\frac{U_{t+1}^s}{\mathcal{R}_t \left(U_{t+1}^s \right)} \right)^{\frac{1}{\psi} - \gamma}$$

Prices of risk

- **Innovation to the log SDF**

$$m_{t+1} - E_t m_{t+1} = -\lambda_z \, \epsilon_{z,t+1} - \lambda_o \, \epsilon_{o,t+1} - \lambda_\eta \, \eta_{o,t+1}$$

$$\lambda_z = \frac{(\gamma - 1/\psi)\, \kappa_1 \sigma_z}{1 - \kappa_1 \rho_z}, \qquad \lambda_o = \frac{(\gamma - 1/\psi)\, \kappa_1 \sigma_o}{1 - \kappa_1 \rho_o}, \qquad \lambda_\eta = \gamma \sigma_\eta$$

- **Permanent components**

$$\ln \frac{\hat{Y}_{t+1}}{\hat{Y}_t} - E\left[\cdot\right] = \frac{\sigma_z}{1 - \rho_z} \, \epsilon_{z,t+1}$$

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Organization capital risk: permanent in the SDF, transitory in consumption

Resolving Permanent Risk Puzzle

- Log Sharpe ratios over long bond

$$\Lambda_\infty = \hat{\lambda}_z, \qquad \Lambda_{max} = \sqrt{\hat{\lambda}_z^2 + \hat{\lambda}_o^2 + \lambda_\eta^2}$$

- How about the R^2?

$$R^2 = \frac{\hat{\lambda}_z^2}{\hat{\lambda}_z^2 + \hat{\lambda}_o^2 + \lambda_\eta^2}$$

OLG with Organization Capital Risk: GE with Production

Fixed physical capital

Production

- Firm i's production function has constant returns to scale in o and k

$$y_{i,t} = \frac{1}{1-\alpha_i} Z_t \, o_{i,t}^{\alpha_i} k_{i,t}^{1-\alpha_i}$$

- α_i is drawn at entry and kept for life, with distribution $F(\alpha)$

Endowment

- Cohorts born with $o_{s,s} = x_0$ and $\bar{k}$ (physical capital)
- Firms rent physical capital from agents in a competitive rental market

Rental market and profits

- Capital demand of firm i: $k_{i,t} = o_{i,t}\, m_t^{-1/\alpha_i}$
- Physical capital is in fixed supply $\bar{k}$ and is rented at $Z_t m_t$ per unit
- Market clearing determines the rent

$$\bar{k} = O_t \int m_t^{-1/\alpha}\, dF(\alpha) \quad \Rightarrow \quad m_t = m(O_t)$$

- Profit is what is left after paying for physical capital

$$\pi_{i,t} = y_{i,t} - Z_t m_t k_{i,t} = \frac{\alpha_i}{1-\alpha_i}\, Z_t\, o_{i,t}\, m_t^{1-1/\alpha_i}$$

- Output is profit plus rent

$$Y_t = Z_t \left[O_t\, \phi(m_t) + \bar{k}\, m_t \right] \equiv Z_t\, y(O_t)$$

where $\phi(m) \equiv \int \frac{\alpha}{1-\alpha}\, m^{1-1/\alpha}\, dF(\alpha)$

Exposure and the O/K sort

- Provided MPK $m_t < 1$, high-α_i firms have a higher O/K ratio
  - $o_{i,t}/k_{i,t} = m_t^{1/\alpha_i}$
- $\uparrow O_t$, $\uparrow$ rent, less so for high-α_i firms. So high-α_i firms benefit more

$$\Delta \ln \pi_{i,t+1} = \Delta \ln Z_{t+1} + \Delta \ln A_{t+1} - \frac{1-\alpha_i}{\alpha_i} \Delta \ln m_{t+1}$$

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- High O/K quintile is the most exposed to organization capital risk
  - Return difference of high minus low O/K carries information on O shocks
  - High O/K has a higher risk premium

Catching up with the frontier

- In real life, organizations can restart if they turn out to be too bad
- For tractability, assume a lower barrier $\bar{O}$ on aggregate organization capital
- If pre-catch-up level $\kappa x_0 + (1-\kappa)\frac{A_t}{A_{t-1}}O_{t-1} < \bar{O}$, every incumbent's organization capital is scaled up by the common factor J_t that just restores the barrier $O_t = \bar{O}$
- Effective growth index of organization capital $\frac{A^c_{t+1}}{A^c_t} = G^c_{t+1} \equiv J_{t+1}G_{t+1}$
- Upgrade is common to every incumbent

Catching up with the frontier

- Incumbents hold a put option: the economy is more risky when O is large
- PD ratio is low when O is large (**duration effect**)

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Predictability:

- PD ratio negatively predicts the market return
- O positively predicts the market return

Preferences and market

- Preferences and the Arrow–Debreu market are the same as in the baseline
- Lifetime budget constraint of cohort s, now with profit and rent as income

$$E_s \sum_{a \geq 0} (1-\kappa)^a \frac{S_{s+a}}{S_s} C^s_{s+a} \leq E_s \sum_{a \geq 0} (1-\kappa)^a \frac{S_{s+a}}{S_s} Z_{s+a} \left[o_{s,s+a} \phi\left(m_{s+a}\right) + \bar{k} m_{s+a} \right]$$

Competitive Equilibrium

An SDF $\{S_t\}$ and consumption plans $\{C^s_t\}$ such that

1. Every generation maximizes U^s_s subject to its lifetime budget constraint;
2. Goods market clears: $\sum_{s \leq t} \kappa (1-\kappa)^{t-s} C^s_t = Y_t = Z_t \, y\left(O_t\right)$

Recursive formulation: unit values and wealth

- $Z_t v^o(X_t)$: per unit value of organization capital
- $Z_t v^k(X_t)$: per unit value of physical capital

$$v^o(X_t) = \phi(m_t) + E_t\left[(1-\kappa)\frac{S_{t+1}}{S_t}e^{\Delta \ln Z_{t+1}} G^c_{t+1} v^o(X_{t+1})\right]$$

$$v^k(X_t) = m_t + E_t\left[(1-\kappa)\frac{S_{t+1}}{S_t}e^{\Delta \ln Z_{t+1}} v^k(X_{t+1})\right]$$

- Wealth
  - Living entrepreneurs: $Z_t\left[O_t v^o(X_t) + \bar{k} v^k(X_t)\right]$
  - Entrants: $Z_t \kappa\left[x_0 v^o(X_t) + \bar{k} v^k(X_t)\right]$

Wealth share of entrants and the growth of survivors

- The wealth share of entrants

$$s\left(X_t\right) = \frac{\kappa\left[x_0\, v^o\left(X_t\right) + \bar{k}\, v^k\left(X_t\right)\right]}{O_t\, v^o\left(X_t\right) + \bar{k}\, v^k\left(X_t\right)}$$

- Without physical capital: $s = \kappa x_0 / O_t \rightarrow$ autarky
- Consumption growth of survivors

$$g\left(X_t, X_{t+1}\right) = \Delta \ln Z_{t+1} + \ln \frac{y\left(O_{t+1}\right)\left[1 - s\left(X_{t+1}\right)\right]}{(1-\kappa)} - \ln y\left(O_t\right)$$

- The SDF

$$\frac{S_{t+1}}{S_t} = \beta\, e^{-\frac{1}{\psi} g_{t+1}} \left(\frac{e^{g_{t+1}} u\left(X_{t+1}\right)}{\mathcal{R}_t\left(e^{g_{t+1}} u\left(X_{t+1}\right)\right)}\right)^{\frac{1}{\psi} - \gamma}$$

Recursive equilibrium

Definition: Recursive Equilibrium

State $X = \{x_z, x_o, O\}$; functions $s(X)$, $v^o(X)$, $v^k(X)$ jointly solve

1. Two pricing equations
2. The wealth share of entrants

A fixed point (algorithm)

$$s \longrightarrow g \longrightarrow (u, S) \longrightarrow \left(v^o, v^k\right) \longrightarrow s$$

Testable Implications

1. Two Factors (long-term growth and organization capital) can span the permanent component of the SDF

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3. Cross-sectional Predictability
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   - Value firm have higher return than growth
4. Time-series Predictability
   - PD ratio negatively predict the market return
   - O positively predict the market return

Quantitative Implication and Empirical Test

- Hypothesis: permanent component of the SDF is spanned by two shocks

$$\log \frac{\hat{S}_{t+1}}{\hat{S}_t} = \alpha - b_g \, g_{t+1} - b_u \, u_{t+1}$$

where $g_{t+1} \equiv r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}$ and u is a proxy for the organization capital shock

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- Transitory-risk adjusted pricing equation

$$E\left(r_j - r_f\right) + \tfrac{1}{2}\mathrm{Var}\left(r_j\right) - \mathrm{Cov}\left(r_j, r^{\infty}_B\right) = \lambda_g \beta_{j,g} + \lambda_u \beta_{j,u}$$

- Factor u is orthogonalized on long equity and the long bond

Data

Proxies for the organization capital shock u

- **Rel. size**: median enterprise value (market cap. + debt) of all firms relative to young firms (Compustat age < 3 years); innovations used as the shock
- **OMK**: high minus low O/K quintile (Eisfeldt and Papanikolaou, 2013)
- **Market**: market log return
- **HML**: value minus growth

Sample

- 1986Q1–2024Q4, quarterly
- Test assets: 25 size–B/M, 10 momentum, 5 O/K, long equity

Prices of Permanent risk

	Growth only	Rel. size	OMK	Market	HML
λ_g	0.38***	0.29**	0.30**	0.21**	0.28***
	(3.56)	(2.16)	(2.04)	(2.13)	(2.61)
λ_u		1.12**	0.70***	0.30***	0.20**
		(2.32)	(4.02)	(3.11)	(2.04)
SSQE	33.05	13.12	20.71	15.41	23.63
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- Both factors carry a positive and significant permanent risk price
- Compared with growth only, adding u cuts pricing errors by 29–60%

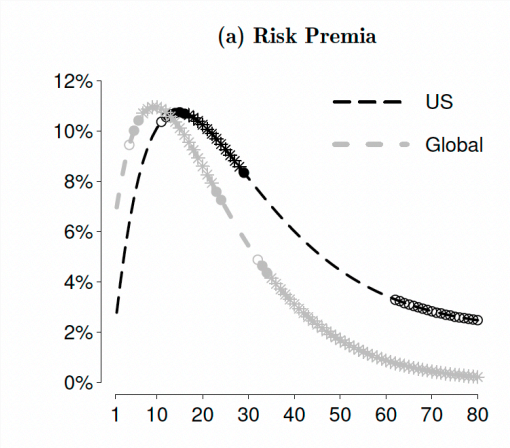
2. Hump-shaped Equity Term Structure

	Dividend strips, maturity in years							Market − 20Y
	1	2	3	4	5	6	7	
Panel A: Data								
Mean	1.79	3.29	4.27	4.78	5.91	6.27	6.59	2.86
Std. dev.	6.59	10.39	10.88	12.30	13.52	14.46	15.38	18.89
Panel B: Model, levered dividend $D_t = Z_t\, y\, (O_t)^3$								
Mean	0.58	2.31	3.70	4.78	5.60	6.20	6.59	3.15
Std. dev.	5.54	6.66	8.84	10.94	12.69	14.04	15.01	6.96

- Duration-matched excess return
- Hump shape is consistent with Binsbergen and Koijen (2017) and Gonçalves (2021)

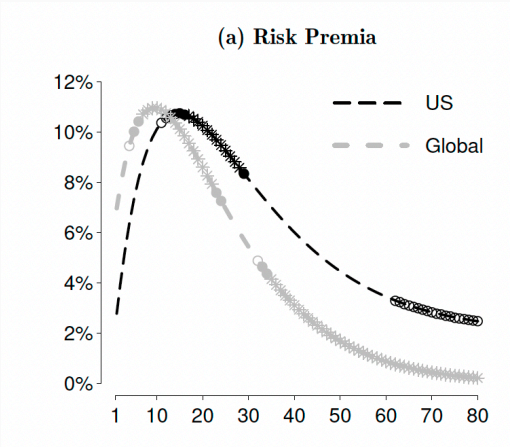
Term structure of equity risk premia

Data: Gonçalves (2021)

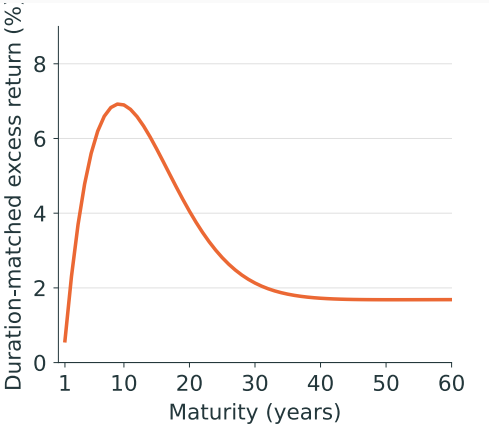


Term structure of equity risk premia

Data: Gonçalves (2021)



My model



3. OMK and Value-Growth Premia

	OMK High − Low O/K	Value − Growth High − Low B/M
Panel A: Data		
Mean	2.70	3.19
Std. dev.	9.55	13.66
Panel B: Model		
Mean	0.69	
Std. dev.	3.31	

4. Predictability of Market Returns

	$\log P/D$		Rel. size ($\log O_t$)	
Horizon	Slope	R^2	Slope	R^2
Panel A: Data, 1986 to 2024, quarterly overlapping				
1 year	-4.80^{**}	0.08	4.02^{**}	0.06
3 years	-11.53^{*}	0.18	11.35^{**}	0.18
5 years	-18.85^{**}	0.32	16.39^{***}	0.24
Panel B: Model, levered dividend $D_t = Z_t\, y\,(O_t)^3$				
1 year	-1.31	0.06	1.45	0.08
3 years	-3.81	0.14	4.21	0.17
5 years	-6.14	0.18	6.79	0.22

- Predictors standardized to unit variance
- Model reports population moments

Wrap up

Conclusion

- A large fraction of permanent risk is orthogonal to long-term growth

Wrap up

Conclusion

- A large fraction of permanent risk is orthogonal to long-term growth
- Long-term shocks to the organization capital of existing cohorts can account for the missing component

Thank you!

Backup: organization capital (Eisfeldt and Papanikolaou, 2013)

- SG&A: selling, general and administrative expenses (Compustat `xsga`)
- Perpetual-inventory method on SG&A expenses

$$O_{i,t} = (1 - \delta_O)\, O_{i,t-1} + \frac{\mathrm{SG\&A}_{i,t}}{\mathrm{cpi}_t}$$

$$O_{i,0} = \frac{\mathrm{SG\&A}_{i,1}/\mathrm{cpi}_1}{g + \delta_O}$$

- $\delta_O = 15\%$, $g = 10\%$; the recursion starts at the firm's first Compustat year
- O/K: organization capital over book assets